RAPID QUALITY EVALUATION AND MARKET VALUATION OF CRUSH TEAR CURL BLACK TEA USING NEAR INFRARED SPECTROSCOPY

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Abstract:

The quality and market valuation of Crush Tear Curl (CTC) tea are traditionally conducted through professional sensory evaluation. A panel of tea tasters assigns a score ranging from 1 to 10, with higher scores signifying better quality and commanding higher prices. The valuation process hinges on three primary attributes collectively referred to as the LIQ value (liquor, infusion and quality), which is influenced by two major biomarkers present in black tea, theaflavins (TF) and thearubigins (TR). Theaflavin is responsible for the colour, brightness and briskness and directly influence the tasters' palate. Thus, combining TF estimation with tasters' scores offers a balanced approach to tea valuation. Human taste perceptions are inherently subjective and may be prone to inconsistencies, while chemical methods, though precise, require costly instruments, skill and are time consuming, thus do not qualify for routine commercial quality estimation [1,2].

In this paper, an attempt has been made to simultaneously predict the LIQ values and TF contents of commercial CTC black tea using near infrared spectroscopy for rapid valuation and decide the market price. For the study, 100 CTC black tea samples were gathered from local markets of Kolkata and Upper Assam. A taster panel evaluated the reference LIQ scores, while TF values were derived by standard spectrophotometric method. The NIR spectra of samples were obtained using a portable NIR spectrometer (Ayudyog Pvt. Ltd., India) with an InGaAs detector, operating in a diffuse reflectance mode [3]. 2 g of each tea powder was placed a quartz sample holder, and spectral scanning was performed over the wavelength range of 900–1700 nm with 16 scans.

Partial Least Square Regression (PLSR) model was employed to predict the component contents from spectra for sensory scores and TF values. To improve the performance of the prediction models, different spectral preprocessing methods including first derivative (1D), second derivative (2D), standard normal variate (SNV) and multiplicative scatter correction (MSC) were applied [3]. Two variable selection methods, namely random frog (RF) and competitive adaptive reweighted sampling (CARS), were applied to select characteristic wavelengths and eliminate irreverent wavelengths. The optimal spectral preprocessing methods for prediction models of L, I, Q and TF were SNV+1D, MSC, SNV+2D, SNV respectively.

The CARS–PLSR model for TF, liquor (L) and quality (Q) shows preferable predictive performance with higher RPD values, lower RMSEP compared with RF–PLSR model; while the RF–PLSR shows better performance for infusion (I). Notably, the CARS -PLSR model for TF yielded the optimal predictive performance, with the highest Rp of 0.983, and the highest RPD value of 5.39 compared to other models after using variable selection algorithms. Overall, the number of wavelengths used for the model establishment represents a reduction above 80%. The results show that the PLSR model using

variable selection methods offers better results than the full spectra model, in terms of Rc, Rp, RMSEC, REMSEP, and RPD. The proposed method is feasible for rapid quality control and taste-related attributes evaluation and valuation of CTC tea.

Keywords: Theaflavin, liquor-infusion-quality (LIQ), RF, CARS

Reference:

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